Lecture 13: Equivalent Resistances and Linearity

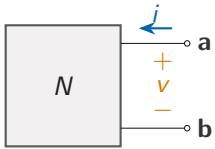
EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

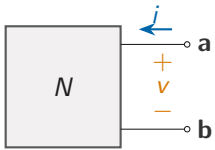
current voltage, node voltage e Δ Y

Review: equivalent one-ports



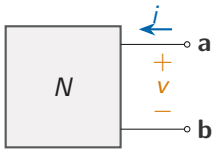
- One-port: a port i, v

Review: equivalent one-ports



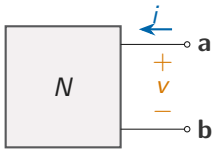
- One-port: a port i, v
- Equivalent: the same i–v characteristic

Review: equivalent one-ports



- One-port: a port i, v
- Equivalent: the same i–v characteristic
- Series, parallel
- Source transformation

Review: equivalent one-ports



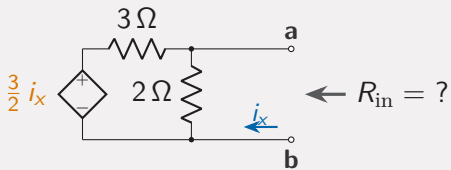
- One-port: a port i, v
- Equivalent: the same i–v characteristic
- Series, parallel
- Source transformation

Node and mesh analysis: the original circuit

Equivalents: a simpler circuit

Motivation

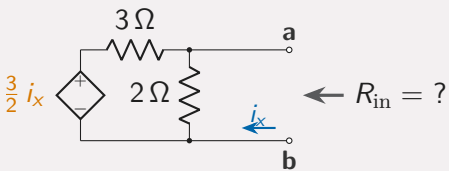
Dependent sources



Series, parallel: not enough

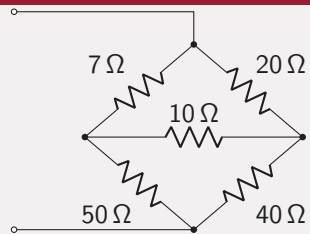
Motivation

Dependent sources



Series, parallel: not enough

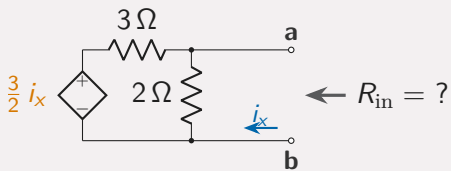
A bridge



No series, no parallel

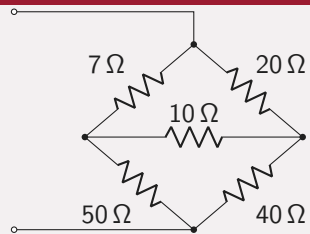
Motivation

Dependent sources



Series, parallel: not enough

A bridge

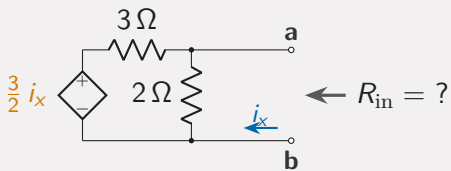


No series, no parallel

One equivalent resistor in both?

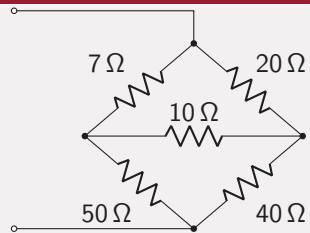
Motivation

Dependent sources



Series, parallel: not enough

A bridge



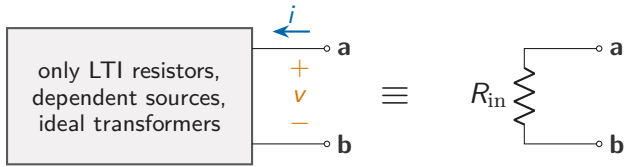
No series, no parallel

One equivalent resistor in both? Yes

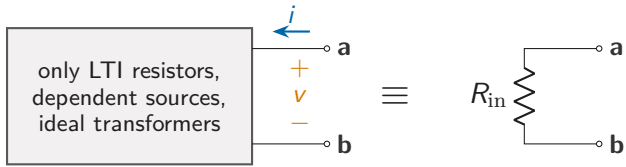
Roadmap


R_{in}
u
y
Input resistance
Δ and Y
Linearity
Superposition
1
2
3
4
a test source
with a dependent source
Δ to Y
Y to Δ
a bridge
$y = \mathbf{k}\,\mathbf{u}$
the implication
one source at a time
three experiments

One-ports without independent sources

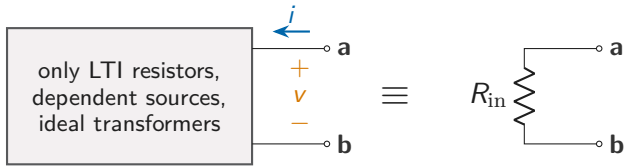


One-ports without independent sources



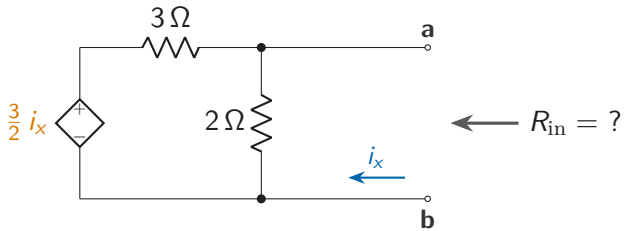
- Controls of the dependent sources: inside the one-port

One-ports without independent sources

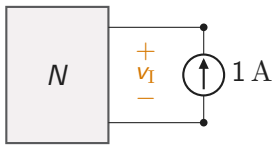


- Controls of the dependent sources: inside the one-port
- $v = R_{\text{in}}\, i$: R_{in} is the input resistance

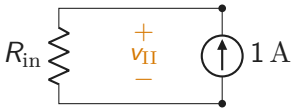
Example 13.1: an input resistance



The idea: a test source

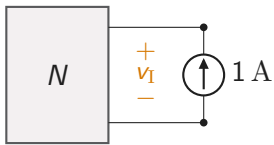


(I)

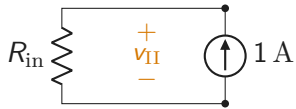


(II)

The idea: a test source



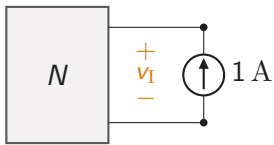
(I)



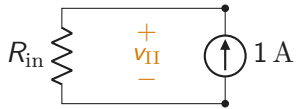
(II)

$$N \equiv R_{\mathrm{in}} \;\Rightarrow\; v_{\mathrm{I}} = v_{\mathrm{II}} = R_{\mathrm{in}} \cdot 1\,\mathrm{A}$$

The idea: a test source



(I)

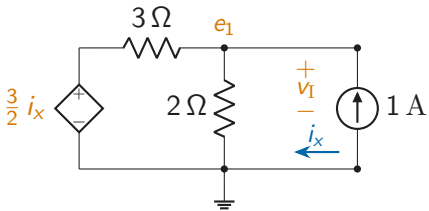


(II)

$$N \equiv R_{in} \;\Rightarrow\; v_I = v_{II} = R_{in} \cdot 1\,\mathrm{A}$$

$$R_{in} = \frac{v_I}{1\,\mathrm{A}}\text{: solve circuit (I)}$$

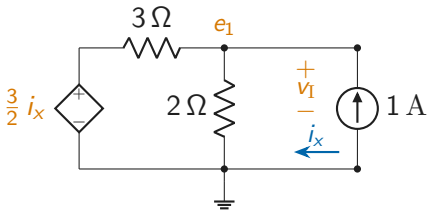
Example 13.1: the solution



Node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

Example 13.1: the solution

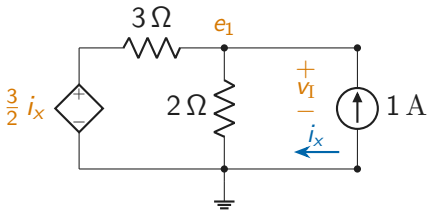


Node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

$i_x = ?$

Example 13.1: the solution



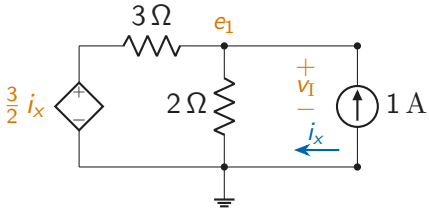
Node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

$i_x = ?$

$i_x = -1\,\text{A}$

Example 13.1: the solution



Node 1:

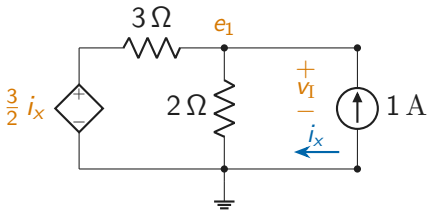
$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

$i_x = ?$

$i_x = -1\,\text{A}$

$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; e_1 = \tfrac{3}{5}\,\text{V}$$

Example 13.1: the solution



Node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

$i_x = ?$

$i_x = -1\,\mathrm{A}$

$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; e_1 = \tfrac{3}{5}\,\mathrm{V}$$

$$R_\mathrm{in} = \frac{v_\mathrm{I}}{1\,\mathrm{A}} = \frac{3}{5}\,\Omega$$

Roadmap









Input resistance | **Δ and Y** | **Linearity** | **Superposition**

✓ | 2 | 3 | 4

a test source ✓

with a dependent source ✓

Δ to Y

Y to Δ

a bridge

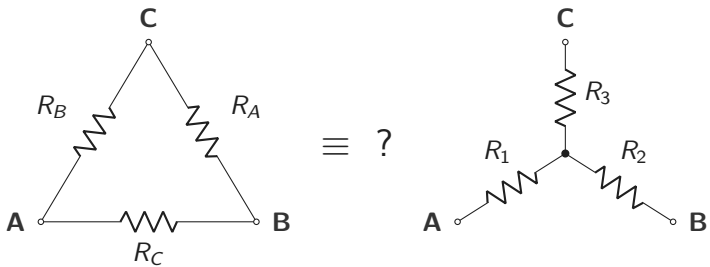
$y = \mathbf{k}\,\mathbf{u}$

the implication

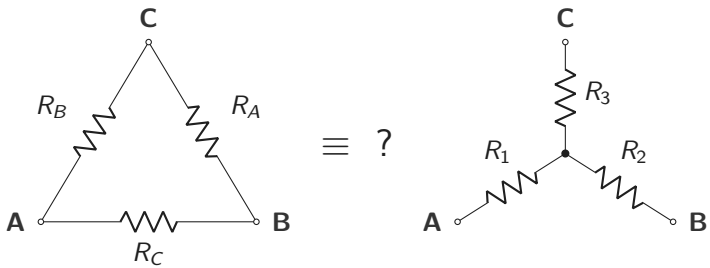
one source at a time

three experiments

Three-terminal circuits: Δ and Y

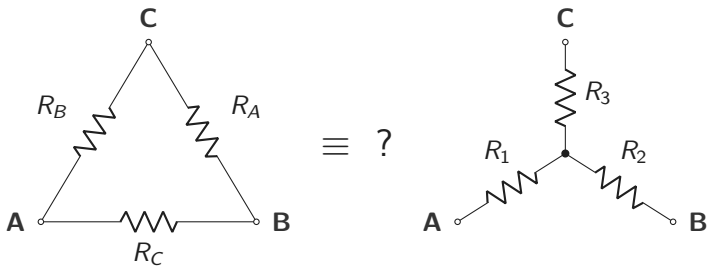


Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

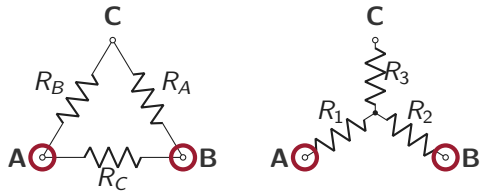
Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

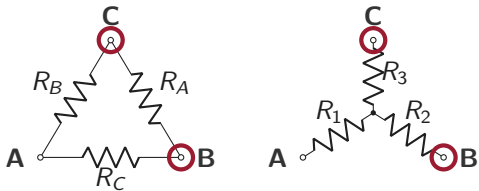
The same resistance between every two terminals

Δ to Y: the resistance between two terminals



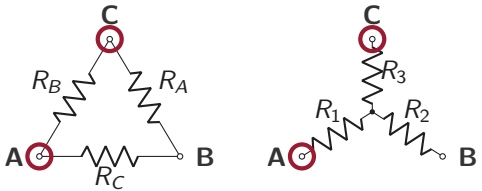
$$\begin{array}{lccc} & \Delta & & \mathrm{Y} \\ \text{A–B:} & \dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C} & = & R_1+R_2 \quad (1) \end{array}$$

Δ to Y: the resistance between two terminals



$$\begin{array}{lccc} & \Delta & & \mathrm{Y} \\ \text{A–B:} & \dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C} & = & R_1+R_2 \quad (1) \\ \text{B–C:} & \dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C} & = & R_2+R_3 \quad (2) \end{array}$$

Δ to Y: the resistance between two terminals



	Δ		Y	
A–B:	$\dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C}$	$=$	R_1+R_2	(1)
B–C:	$\dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C}$	$=$	R_2+R_3	(2)
C–A:	$\dfrac{R_B(R_C+R_A)}{R_A+R_B+R_C}$	$=$	R_3+R_1	(3)

Δ to Y: the result

$$\frac{1}{2}\left[(1)+(2)+(3)\right]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \tag{4}$$

Δ to Y: the result

$$\tfrac{1}{2}\big[(1)+(2)+(3)\big]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2)$$

Δ to Y: the result

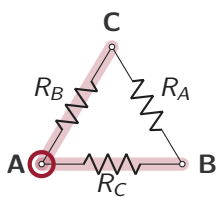
$$\tfrac{1}{2}\big[(1)+(2)+(3)\big]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2) \qquad R_2=\frac{R_CR_A}{R_A+R_B+R_C} \qquad R_3=\frac{R_AR_B}{R_A+R_B+R_C}$$

Δ to Y: the result

$$\tfrac{1}{2}\big[(1)+(2)+(3)\big]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2) \qquad R_2=\frac{R_CR_A}{R_A+R_B+R_C} \qquad R_3=\frac{R_AR_B}{R_A+R_B+R_C}$$



Y resistor at a terminal = product of the two Δ resistors there / sum of the Δ resistors

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\left(R_A + R_B + R_C\right)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

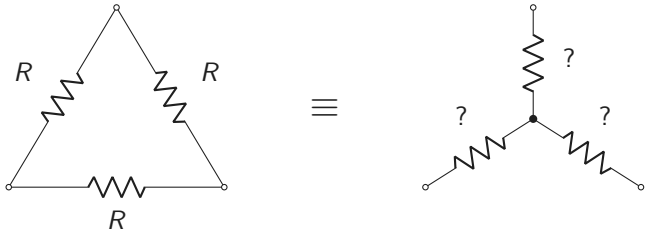
$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

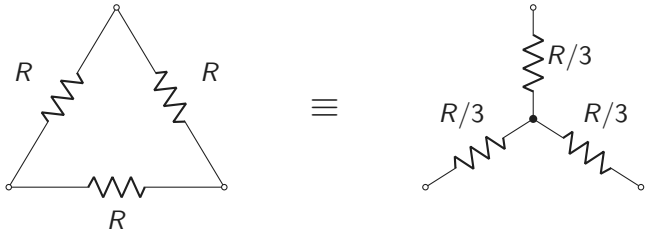
$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

$$\Delta \text{ resistor} = \frac{\text{sum of the products of the Y resistors}}{\text{the Y resistor opposite}}$$

Equal resistors

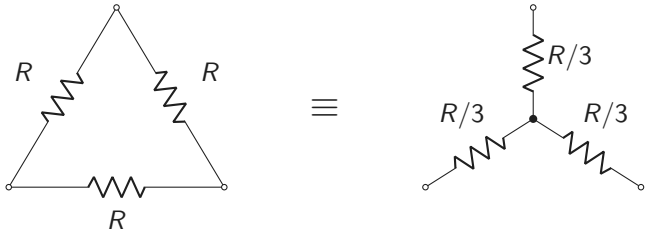


Equal resistors



$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

Equal resistors



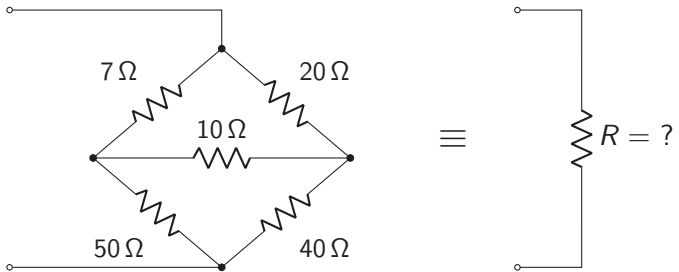
$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

Y with $R \equiv \Delta$ with $3R$

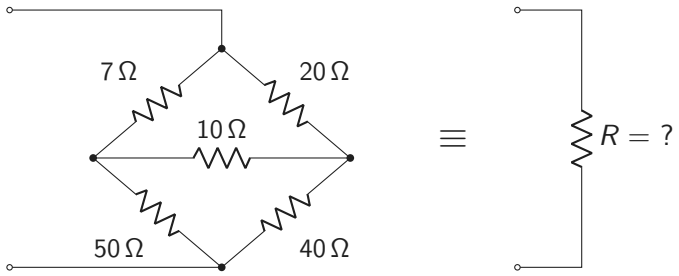
Roadmap


R_{in}
u
y
Input resistance
Δ and Y
Linearity
Superposition
✓
2
3
4
a test source ✓
with a dependent source ✓
Δ to Y ✓
Y to Δ ✓
a bridge
$y = \mathbf{k}\,\mathbf{u}$
the implication
one source at a time
three experiments

Example 13.2: a bridge

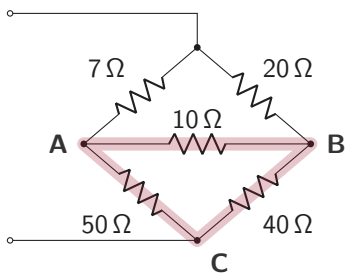


Example 13.2: a bridge



No two resistors in series or in parallel

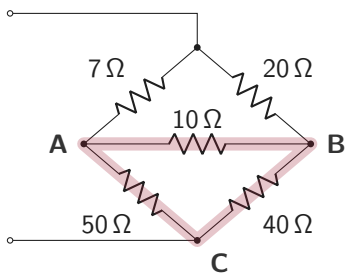
Example 13.2: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

Example 13.2: Δ to Y

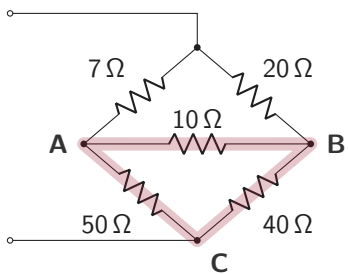


Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

$R_A + R_B + R_C = 100\,\Omega$

Example 13.2: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

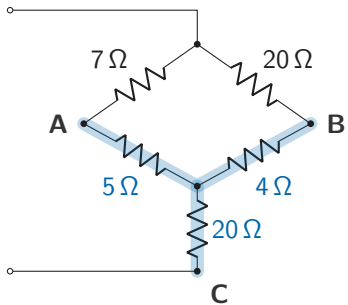
$R_A + R_B + R_C = 100\,\Omega$

$$R_1 = \frac{50 \cdot 10}{100} = 5\,\Omega$$

$$R_2 = \frac{10 \cdot 40}{100} = 4\,\Omega$$

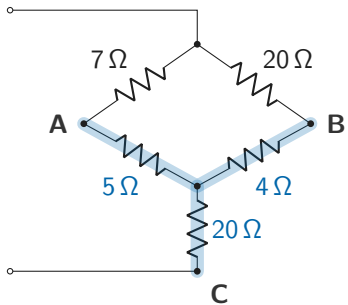
$$R_3 = \frac{40 \cdot 50}{100} = 20\,\Omega$$

Example 13.2: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

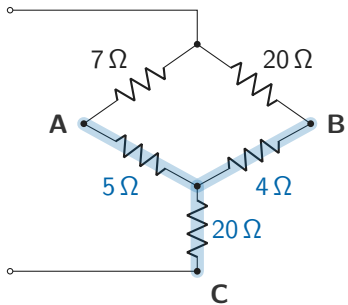
Example 13.2: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Example 13.2: the new circuit



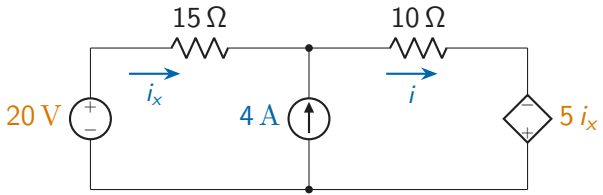
Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Series: $8 + 20$

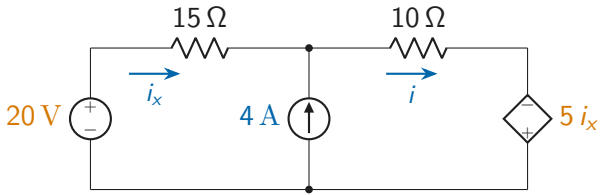
$$R = 28\,\Omega$$

Motivation: many sources



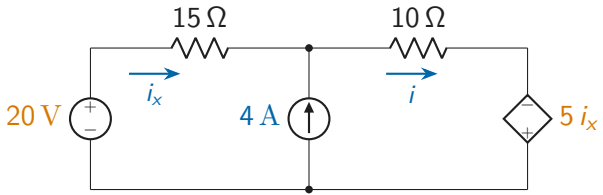
- Two independent sources: the inputs

Motivation: many sources



- Two independent sources: the inputs
- One source at a time: two simpler circuits?

Motivation: many sources

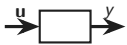


- Two independent sources: the inputs
- One source at a time: two simpler circuits?
- Yes: the circuit is linear

Roadmap









Input resistance ✓

a test source ✓

with a dependent source ✓

Δ and Y ✓

Δ to Y ✓

Y to Δ ✓

a bridge ✓

Linearity 3

$y = \mathbf{k}\,\mathbf{u}$

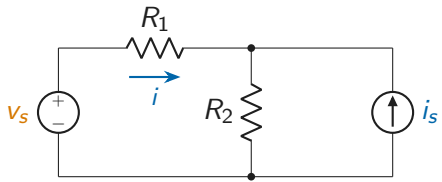
the implication

Superposition 4

one source at a time

three experiments

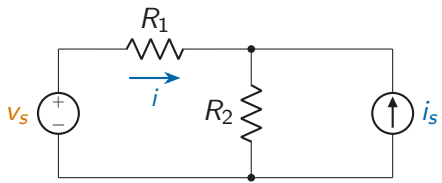
Linearity: inputs and outputs



Input vector: the independent sources

$$\mathbf{u} = \begin{bmatrix} v_s \\ i_s \end{bmatrix}$$

Linearity: inputs and outputs



Input vector: the independent sources

$$\mathbf{u} = \begin{bmatrix} v_s \\ i_s \end{bmatrix}$$

Output y: any branch voltage or current

e.g. $y = i$

Linearity

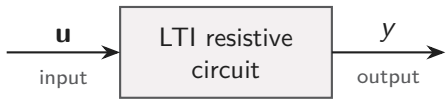
For each output y, there is $\mathbf{k} = [\, k_1 \ \cdots \ k_m \,]$ such that

$$y = \mathbf{k}\,\mathbf{u} = k_1 u_1 + k_2 u_2 + \cdots + k_m u_m$$

Linearity

For each output y, there is $\mathbf{k} = [\, k_1 \ \cdots \ k_m \,]$ such that

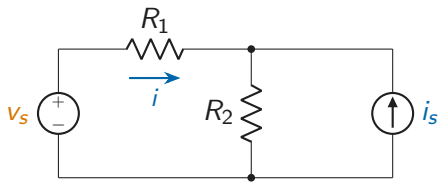
$$y = \mathbf{k}\,\mathbf{u} = k_1 u_1 + k_2 u_2 + \cdots + k_m u_m$$



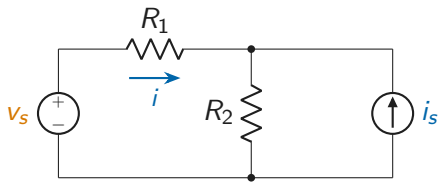
$$y = y(\mathbf{u})$$

Example 13.3: k for $y = i$

$i = k_1 v_s + k_2 i_s$: $k_1, k_2 = ?$



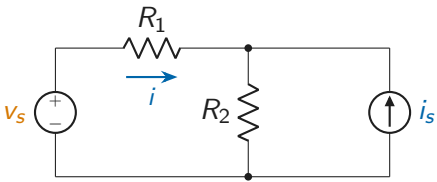
Example 13.3: k for $y = i$



$i = k_1 v_s + k_2 i_s$: $k_1, k_2 = ?$

KVL: $-v_s + R_1 i + R_2(i + i_s) = 0$

Example 13.3: k for $y = i$

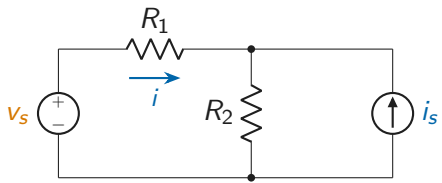


$i = k_1 v_s + k_2 i_s$: $k_1, k_2 = ?$

KVL: $-v_s + R_1 i + R_2(i + i_s) = 0$

$$i = \frac{1}{R_1 + R_2} v_s - \frac{R_2}{R_1 + R_2} i_s$$

Example 13.3: k for $y = i$



$i = k_1 v_s + k_2 i_s$: $k_1, k_2 = ?$

KVL: $-v_s + R_1 i + R_2(i + i_s) = 0$

$$i = \frac{1}{R_1 + R_2} v_s - \frac{R_2}{R_1 + R_2} i_s$$

$$k_1 = \frac{1}{R_1 + R_2}, \qquad k_2 = -\frac{R_2}{R_1 + R_2}$$

Linearity: the implication

$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w})$$

$\mathbf{u}$, $\mathbf{w}$: input vectors; α, β: numbers

Linearity: the implication

$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w})$$

u, **w**: input vectors; α, β: numbers

$$y\left(\begin{bmatrix} v_s \\ i_s \end{bmatrix}\right) = y\left(\begin{bmatrix} v_s \\ 0 \end{bmatrix}\right) + y\left(\begin{bmatrix} 0 \\ i_s \end{bmatrix}\right)$$

Linearity: the implication

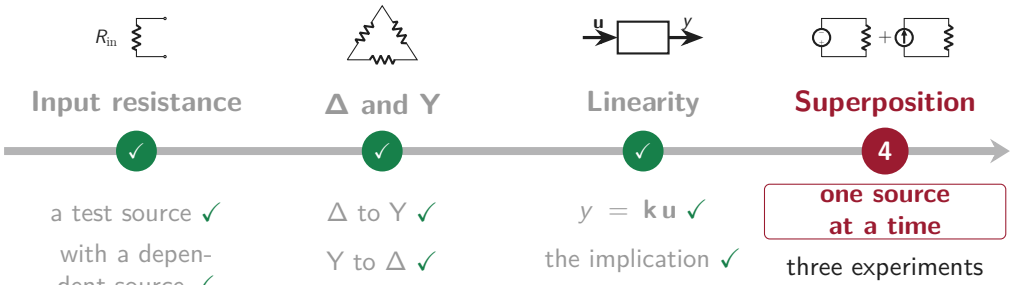
$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w})$$

u, **w**: input vectors; α, β: numbers

$$y\left(\begin{bmatrix} v_s \\ i_s \end{bmatrix}\right) = y\left(\begin{bmatrix} v_s \\ 0 \end{bmatrix}\right) + y\left(\begin{bmatrix} 0 \\ i_s \end{bmatrix}\right)$$

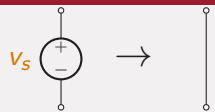
Superposition: one independent source at a time

Roadmap


R_{in}
u
y
Input resistance
Δ and Y
Linearity
Superposition
✓
✓
✓
4
a test source ✓
with a dependent source ✓
Δ to Y ✓
Y to Δ ✓
a bridge ✓
$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓
one source at a time
three experiments

Killing a source

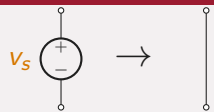
Voltage source



$v_s = 0$: a short circuit

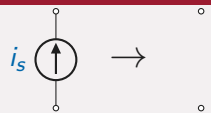
Killing a source

Voltage source



$v_s = 0$: a short circuit

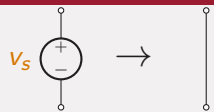
Current source



$i_s = 0$: an open circuit

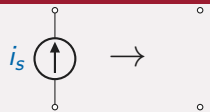
Killing a source

Voltage source



$v_s = 0$: a short circuit

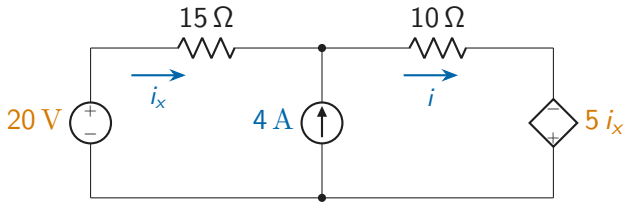
Current source



$i_s = 0$: an open circuit

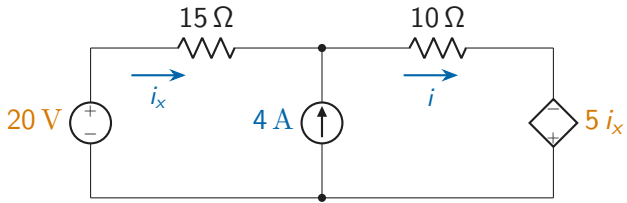
Do not kill dependent sources: they are not inputs

Example 13.4: superposition



Find i, one source at a time

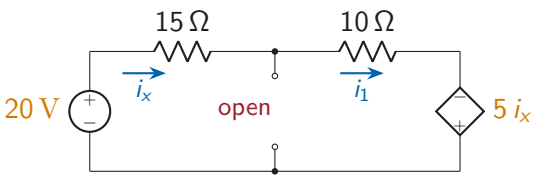
Example 13.4: superposition



Find i, one source at a time

$i = i_1 + i_2$: i_1: only 20 V, i_2: only 4 A

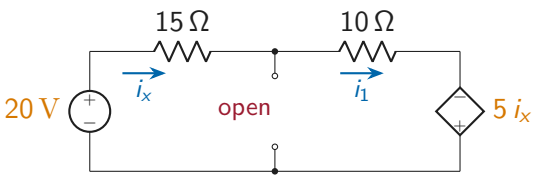
Example 13.4: kill the current source



KVL:

$$-20 + 15\,i_x + 10\,i_1 - 5\,i_x = 0$$

Example 13.4: kill the current source

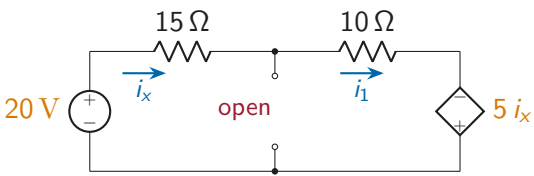


KVL:

$$-20 + 15\, i_x + 10\, i_1 - 5\, i_x = 0$$

$$i_x = ?$$

Example 13.4: kill the current source



KVL:

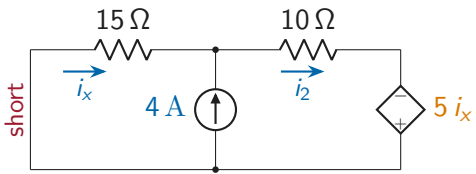
$$-20 + 15\,i_x + 10\,i_1 - 5\,i_x = 0$$

$i_x = ?$

$i_x = i_1 \;\Rightarrow\; 20\,i_1 = 20$

$$\boxed{i_1 = 1\,\mathrm{A}}$$

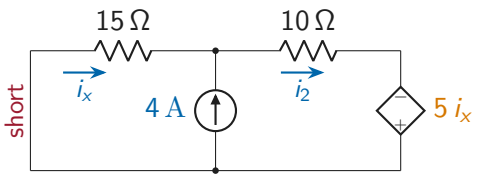
Example 13.4: kill the voltage source



KVL:

$$15\, i_x + 10\, i_2 - 5\, i_x = 0$$

Example 13.4: kill the voltage source

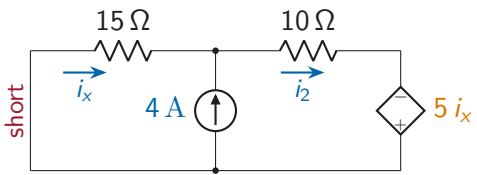


KVL:

$15\, i_x + 10\, i_2 - 5\, i_x = 0$

$i_x = ?$

Example 13.4: kill the voltage source



KVL:

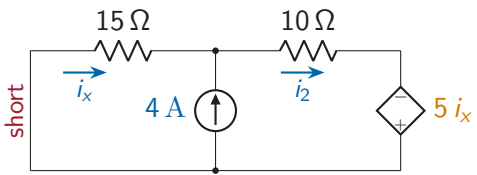
$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

$i_x = i_2 - 4 \;\Rightarrow\; 20\,i_2 = 40$

$i_2 = 2\,\text{A}$

Example 13.4: kill the voltage source



KVL:

$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

$i_x = i_2 - 4 \;\Rightarrow\; 20\,i_2 = 40$

$i_2 = 2\,\mathrm{A}$

$i = i_1 + i_2 = 3\,\mathrm{A}$

Roadmap



Input resistance

✓

a test source ✓

with a dependent source ✓



Δ and Y

✓

Δ to Y ✓

Y to Δ ✓

a bridge ✓

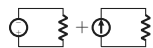


Linearity

✓

$y = \mathbf{k}\,\mathbf{u}$ ✓

the implication ✓



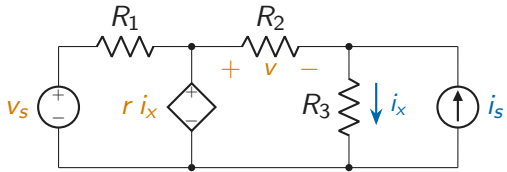
Superposition

4

one source at a time ✓

three experiments

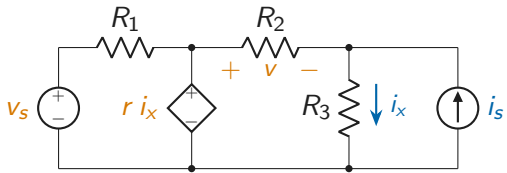
Example 13.5: three experiments



	v_s	i_s	v
Exp. 1	$12\,\mathrm{V}$	0	v_1
Exp. 2	$-6\,\mathrm{V}$	$6\,\mathrm{A}$	v_2
Exp. 3	$4\,\mathrm{V}$	$4\,\mathrm{A}$	v_3

v_3 in terms of v_1 and v_2?

Example 13.5: three experiments



	v_s	i_s	v
Exp. 1	$12\,\mathrm{V}$	0	v_1
Exp. 2	$-6\,\mathrm{V}$	$6\,\mathrm{A}$	v_2
Exp. 3	$4\,\mathrm{V}$	$4\,\mathrm{A}$	v_3

v_3 in terms of v_1 and v_2?

Linearity: $v = \mathbf{k} \begin{bmatrix} v_s \\ i_s \end{bmatrix}$

Example 13.5: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}$$

Example 13.5: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

Example 13.5: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

$$v_3 = v\left(\alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}\right) = \alpha\, v_1 + \beta\, v_2$$

Example 13.5: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

$$v_3 = v\left(\alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}\right) = \alpha\, v_1 + \beta\, v_2$$

$$\boxed{v_3 = \tfrac{2}{3}\, v_1 + \tfrac{2}{3}\, v_2}$$

No element values needed

Summary

In the circuit	What to do
resistors, dependent sources	$R_{\text{in}} = v/1\,\text{A}$ with a $1\,\text{A}$ test source
a Δ	Y: $R_1 = R_B R_C/(R_A + R_B + R_C)$, ...
a Y	Δ: $R_A = (R_1R_2 + R_2R_3 + R_3R_1)/R_1$, ...
equal resistors	Δ with R $\equiv$ Y with $R/3$
linear circuit	$y = \mathbf{k}\,\mathbf{u}$: $y(\alpha\mathbf{u} + \beta\mathbf{w}) = \alpha y(\mathbf{u}) + \beta y(\mathbf{w})$
several sources	superposition: killed v_s = short, killed i_s = open